

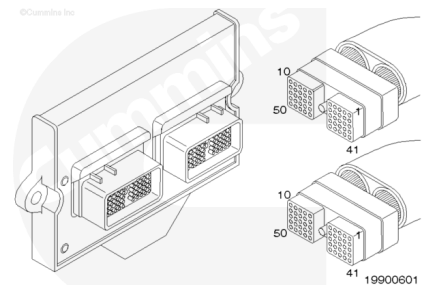
## Trainings ▾

# Pin Replacement

## Deutsch

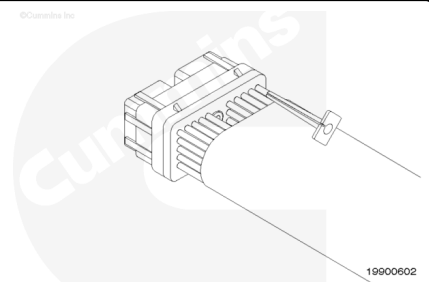
### Deutsch

The 50-pin Deutsch connector is used to connect the OEM and engine harnesses to the ECM.

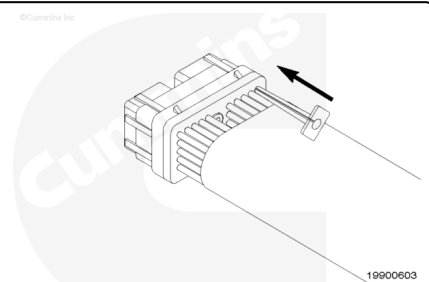


Use Deutsch extraction tool, Part Number 3824815, over the wire to remove a pin from the connector.

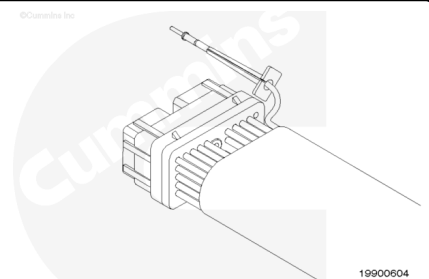
**Note :** Replace one wire at a time. If more than one wire needs replacement, attach a lettered tag to each wire removed. Refer to the wiring diagram in Section E of this manual.



Push the tool into the connector about 25 mm [1 in] until it bottoms on the terminal flange.



Hold the tool on the terminal flange and pull the wire and the connector pin out of the connector. Note and record the hole from which the pin is removed.

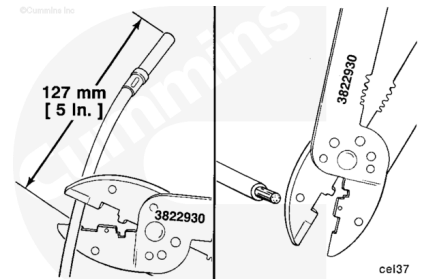


**Note :** The repair wire is 127 mm [5 in] long.

The repair wire for the female terminal is Part Number 3824809.

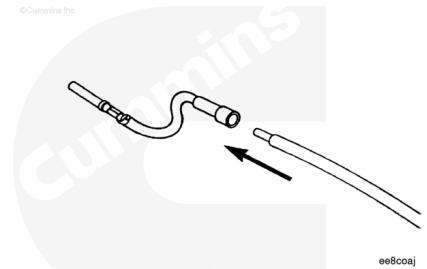
Use wire crimp tool, Part Number 3822930, to cut 127 mm [5 in] off the wire and pin.

Use the crimp tool to remove 6 mm [1/4 in] of insulation from the wire.

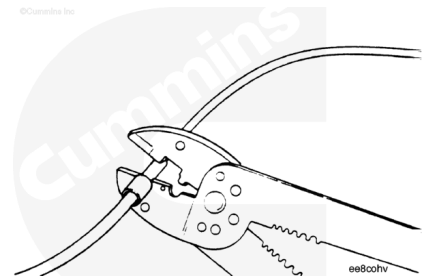


Install the correct repair wire on the bare wire.

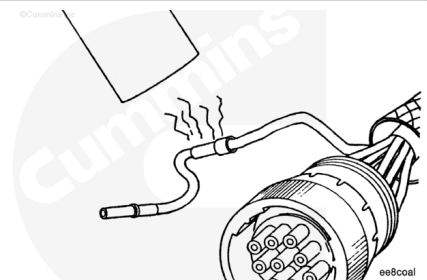
**Note :** Make sure the bare wire extends into the splice connector.



Use the wire crimp tool to crimp the repair wire onto the bare wire.



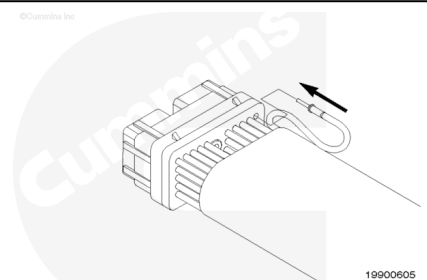
Use a heat gun, Part Number 3822860, to heat the shrink tubing around the wire. The tubing will shrink and make the connection waterproof.



Insert the pin into the correct hole of the connector.

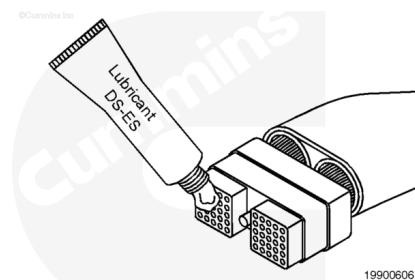
The pin **must** click into place and hold the wire in the connector.

Pull the wire gently to make sure it is seated in the connector.



**⚠ CAUTION ⚠**

Use only Cummins-recommended lubricant DS-ES, Part Number 3822934. Other lubricants, such as lubricating oil or grease, in the connectors can cause ECM damage, poor engine performance, or premature connector pin wear.



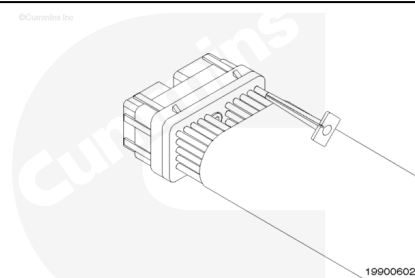
Apply a small amount of lubricant to the connector terminals. Do **not** fill the entire connector cavity with lubricant.

## Connector Replacement

### Deutsch

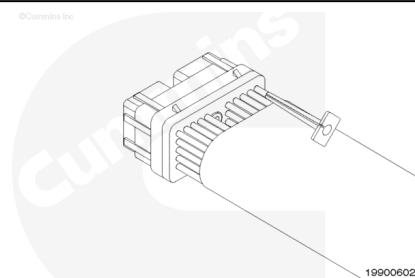
#### Deutsch

The 50-pin Deutsch connector is used to connect the OEM interface and engine harnesses to the ECM.

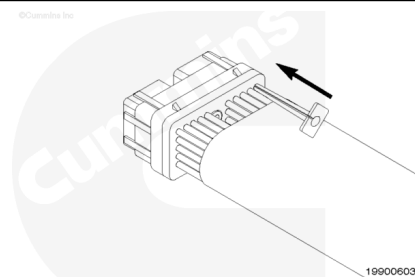


To replace the connector, use Deutsch extraction tool, Part Number 3824815, over each wire to remove all pins from the connector.

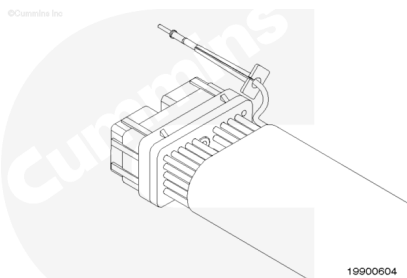
**Note :** Replace one wire at a time. Attach a lettered tag to each wire removed. Refer to the wiring diagram in Section E of this manual.



Push the tool into the connector about 25 mm [1 in] until it bottoms out on the terminal flange.



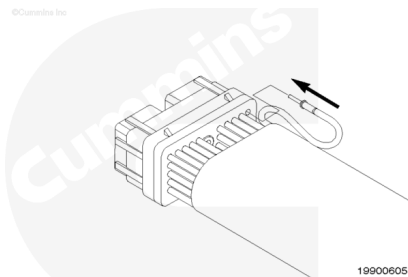
Hold the tool on the terminal flange and pull the wire and the connector pin out of the connector. Note and record the hole from which the pin is removed.



19900604

The replacement OEM harness connector is Part Number 3162732.

The replacement engine harness connector is Part Number 3162733.

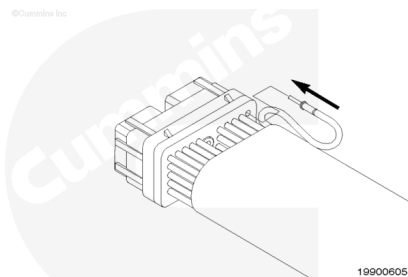


19900605

Insert the pins into the correct holes of the replacement connector.

Each pin **must** click into place and hold the wires in the connector.

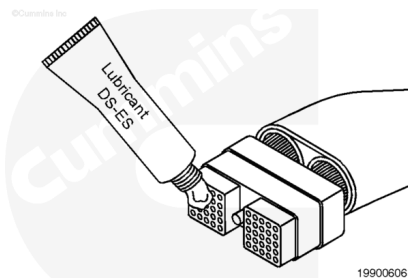
Pull each wire gently to make sure it is seated in the connector.



19900605

### ⚠ CAUTION ⚠

**Use only Cummins-recommended lubricant DS-ES, Part Number 3822934. Other lubricants, such as lubricating oil or grease, in the connectors can cause ECM damage, poor engine performance, or premature connector pin wear.**



19900606

Apply a small amount of lubricant to the connector terminals. Do **not** fill the entire connector cavity with lubricant.